INDEVOLT

OpenData Application Note

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Revision History

Version	Description	Applicable firmware version
V1.0	Initial release	V1.3.07_R006.032_M4848_00000026
V1.1	Restructure the document and update API documentation	● BK1600/BK1600Ultra: V1.3.0A_R006.072_M4848_00000039 ● SolidFlex2000/PowerFlex2000: V1.3.09_R00D.012_M4801_00000015
V1.2	Add curl command example	
V1.3	Add cJSON points for SolidFlex2000 / PowerFlex2000	● BK1600/BK1600Ultra: V1.3.0A_R006.072_M4848_00000039 ● SolidFlex2000/PowerFlex2000: ■ CMS：V1406.07.002B ■ Pfile：V0D.00.11
V1.4	Correct cJSON points for the SolidFlex2000 / PowerFlex2000	

1. Introduction

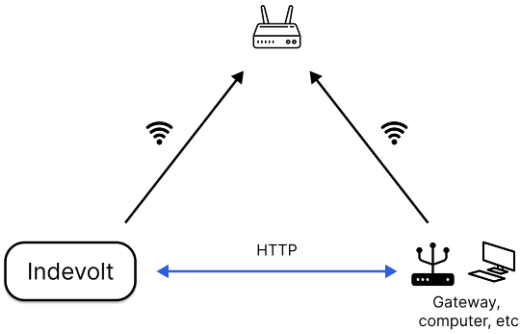
OpenData is a lightweight communication framework designed for WiFi-based INDEVOLT IoT devices. Devices connect to local networks via WiFi, supporting both active data push and response to external queries.

Core Features

- **Device Data Acquisition**: Real-time pulling of device data.
- **Device Control**: Dynamic adjustment of device parameters and configurations.

Use Cases

- Receiving external HTTP query requests.



2. Preparations

Step 1: Install Tools

- Postman / cURL: Used for invoking HTTP APIs to retrieve device data or update device configurations.
- Any network debugging tool (e.g., NetAssist): Used for receiving UDP broadcasts.

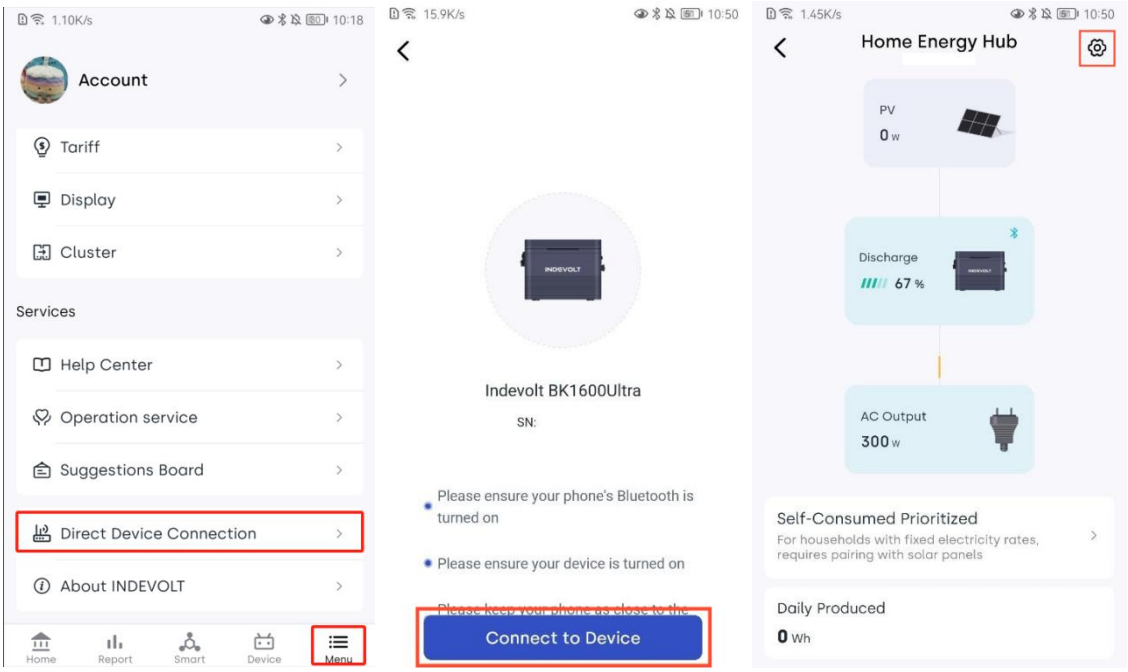
Step 2: Enable API

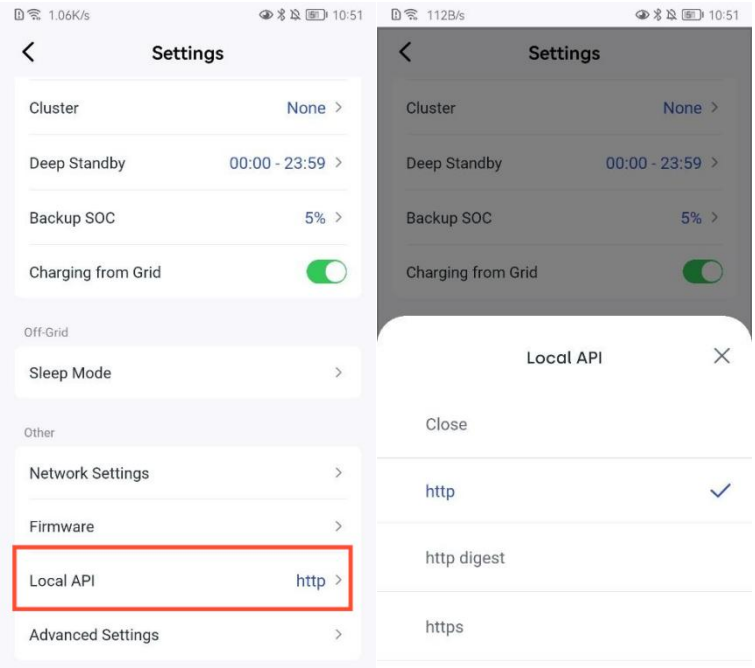
Device APIs are **disabled by default** and must be enabled before use.

OpenData provides three modes:

- HTTP
- HTTP+ Digest Authentication
- HTTPS (Currently unsupported, coming soon)

Steps:

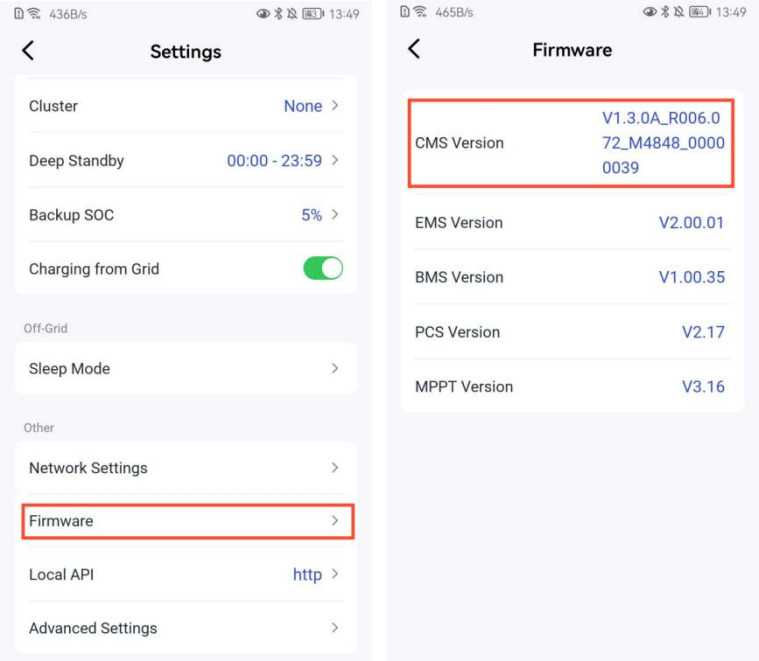




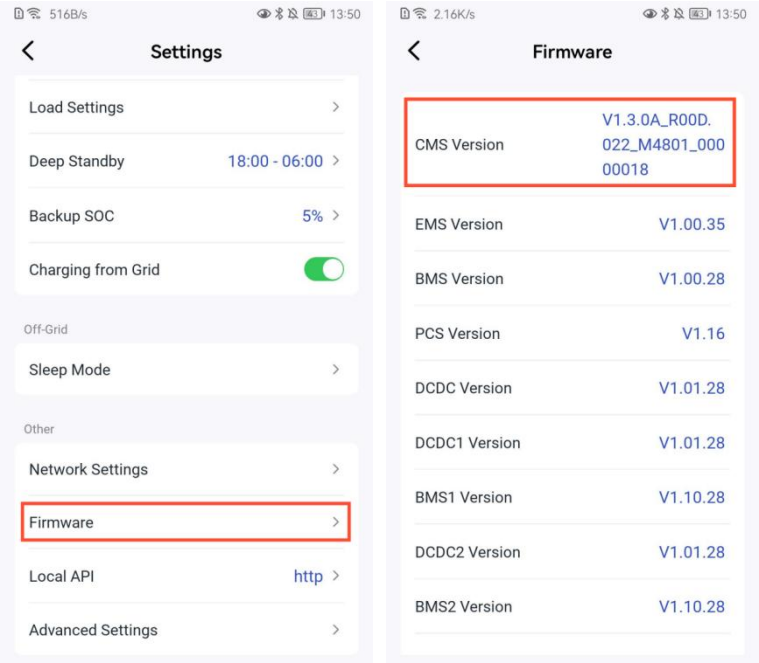
Step 3: Check Firmware Version

Check the device firmware version in the INDEVOLT app. If the version is lower than that specified in the Revision History, update the device firmware.

- INDEVOLT BK1600/BK1600Ultra



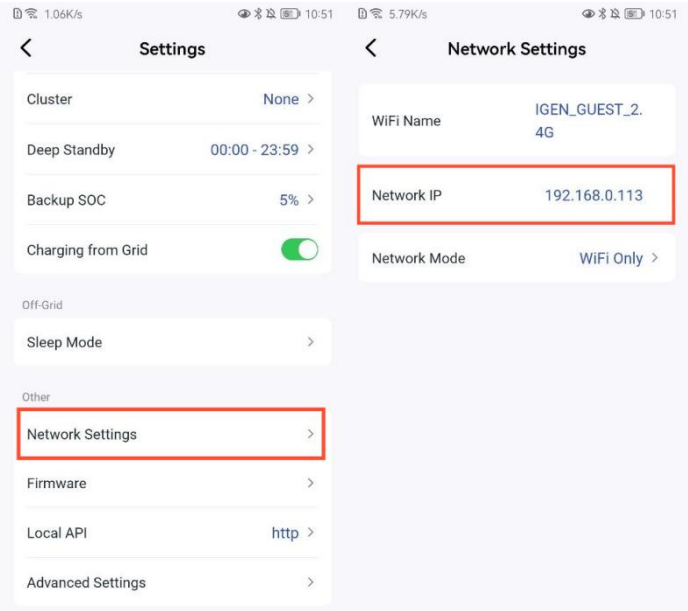
- INDEVOLT SolidFlex2000/PowerFlex2000



Step 4: Obtain IP Address

Choose any one of the following methods:

- Query via router's management list;
- Check in APP device settings:

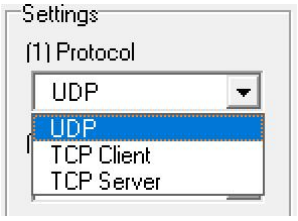


- Obtain IP via UDP broadcast:

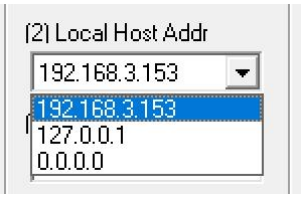
(1) Ensure the device's WiFi network and computer are on the same local area network.

(2) Open a network debugging tool.

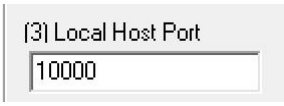
(3) Select **UDP** protocol.



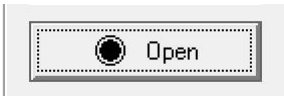
(4) Select **Local Host Addr**.



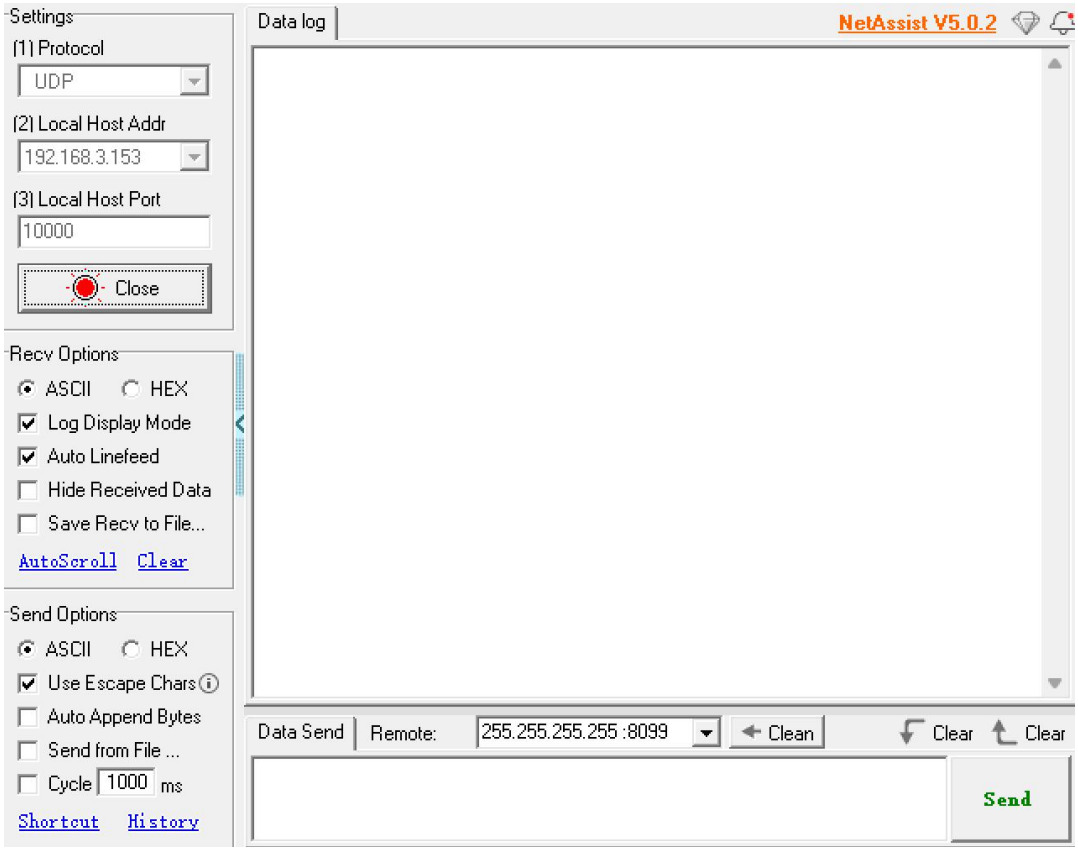
(5) Set **Local Host Post** to **10000.**



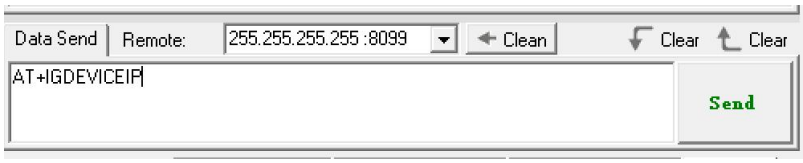
(6) Click **Open**.



(7) Configure Remote with **broadcast address and port**: 255.255.255.255:8099.



(8) Enter AT command in message box: **AT+IGDEVICEIP**.



(9) Click **Send.**

(10) INDEVOLT devices on the same network will respond with their IP address and serial number (SN).

```
[2025-06-26 15:46:36.633]# SEND ASCII TO 255.255.255.255 :8099>
AT+IGDEVICEIP

[2025-06-26 15:46:37.023]# RECV ASCII FROM 192.168.3.119 :8099>
{"sn":"D850BD20A300129S","fw":"V1.3.07_R003.001_MP12W_
00000012","ip":"192.168.3.119","opendata_ver":"V1.0.1"}
```

3. HTTP Usage

Request Structure

Request Methods:

- GET: Requests the server to return specified resources.
- POST: Requests the server to perform specified operations.

Request Address:

```
http://{IP_ADDRESS}:8080/rpc/{API}
```

- {IP_ADDRESS}: Device IP address.
- {API}: HTTP API to be invoked.

Request Example:

- Retrieve device data:

```
POST
http://192.168.31.213:8080/rpc/Indevolt.GetData?config={"t":[1664,1665]}
```

cURL command Example:

- Retrieve battery SOC:

```
curl -g -X POST -H "Content-Type: application/json"
"http://192.168.1.75:8080/rpc/Indevolt.GetData?config={\"t\":[6002]}"
```

Digest Authentication

Digest Authentication technology verifies user identity in network communications, preventing plaintext password transmission.

In HTTP+Digest mode:

- New or factory-reset devices must first use the User.SetConfig interface to modify the default password.
- Other APIs become accessible only after successful password modification.

Required tools:

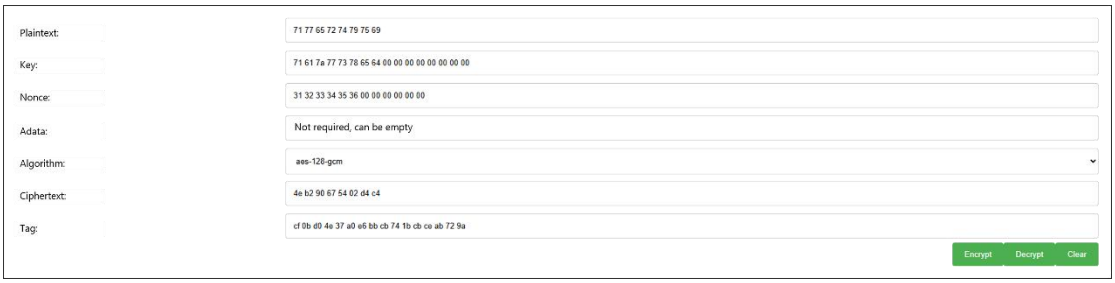
- ASCII string to hexadecimal conversion
- Convert hexadecimal to base64
- AES_GCM encryption tool

Example of modifying password:

1. Convert the new password, original password, and random number from string to hexadecimal.

	ASCII string	**Hexadecimal**
New password	qwertyui	71 77 65 72 74 79 75 69
Original password	qazwsxed	71 61 7a 77 73 78 65 64 00 00 00 00 00 00 00 00 (Supplement 0 to 16 bytes)
Random number	123456	31 32 33 34 35 36 00 00 00 00 00 00 (Supplement 0 to 12 bytes)

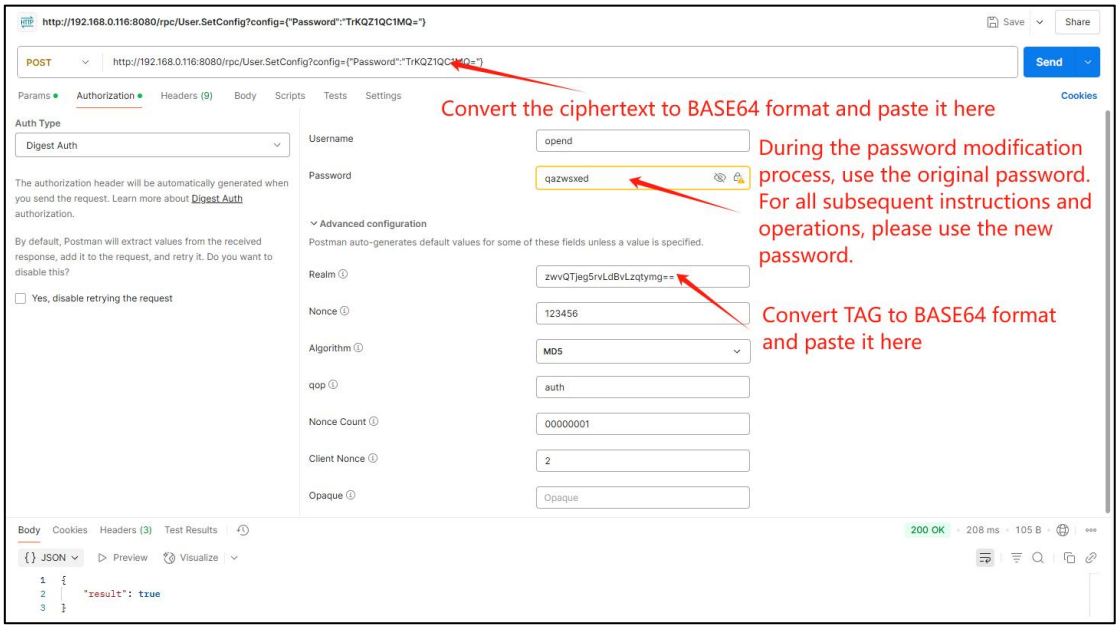
2. Use AES_GCM encryption tool to encrypt, fill in the corresponding information as follows.



3. Convert ciphertext and tag from hexadecimal to base64.

	Hexadecimal	**base64**
Ciphertext	4e b2 90 67 54 02 d4 c4	TrKQZ1QC1MQ=
Tag	cf 0b d0 4e 37 a0 e6 bb cb 74 1b cb ce ab 72 9a	zwvQTjeg5rvLdBvLzqtymg==

4. The result returns true, indicating successful modification.



Parameter	**Type**	**Describe**	**Attribute**
Username	character string	Default "opend"	required
Password	character string	Default device key. ● Using the default password can only request the User.SetConfig interface to change the password. ● Using the modified password can request other interfaces.	required

Realm	character string	● If calling the password modification interface and changing the password, you need to pass the AES128-GCM tag value. ● If calling other interfaces in Base64 format, you can use a random value	required
Nonce	Default type of Digest	Random values can be used	required
Algorithm	Default type of Digest	MD5	required
qop	Default type of Digest	auth	required
Nonce Count	Default type of Digest	Random values can be used	required
Client Nonce	Default type of Digest	Random values can be used	required

Errors

- 400 Bad Request: The server cannot understand the format of the request; the client should modify the request and try again.
- 401 Unauthorized: The request requires authentication; the client needs to provide valid credentials.
- 403 Forbidden: The server understands the request but refuses to execute it, usually due to permission issues.
- 404 Not Found: The server cannot find the requested resource, possibly

because the resource does not exist or has been deleted.

- 405 Method Not Allowed: The requested method is incompatible with the resource, for example, performing a write operation on a read-only resource.
- 408 Request Timeout: The server times out while waiting for the request; the client can retry later.
- 409 Conflict: The request conflicts with the current state of the resource, for example, multiple users editing the same resource simultaneously.
- 410 Gone): The requested resource has been permanently deleted and there is no new address.
- 500 Internal Server Error: The server encounters an unknown error and cannot complete the request.
- 501 Not Implemented: The server does not support the requested method and cannot execute it.
- 502 Bad Gateway: The server, acting as a gateway or proxy, received an invalid response from the upstream server.
- 503 Service Unavailable: The server is currently unable to handle the request, possibly due to overload or maintenance.
- 504 Gateway Timeout: The server, acting as a gateway or proxy, did not receive a timely response from the upstream server.
- 505 HTTP Version Not Supported: The server does not support the HTTP version used in the request.

4. HTTP API

Component	Description
Indevolt	Retrieve device data and control device.
Sys	Obtain device information.

Indevolt

1. Retrieve device data

Request example	**POST** http://192.168.31.213:8080/rpc/Indevolt.GetData?config={"t":[1664,1665]}

Response example	{
	“1664”:100,
	“1665”:251
	}

Function: Indevolt.GetData

Parameter

Parameter	Type	Description
config	Object	t: The request address for data, specified in the **cJSON Point** column of the table below.

Return value

Device data in JSON format.

- **INDEVOLT BK1600/BK1600Ultra**

cJSON Point	cJSON Value Type	Unit	Point Description	Enum Definition
0	Float		Device SN	
7101	Enum		Working mode	0: Outdoor Portable 1: Self-consumed Prioritized 5: Charge/Discharge Schedule
1664	Float	W	DC Input Power1	
1665	Float	W	DC Input Power2	
2108	Float	W	Total AC Output Power	
1502	Float	kWh	Daily Production	
1505	Float	0.001kWh	Cumulative Production	
2101	Float	W	Total AC Input Power	
2107	Float	kWh	Total AC Input Energy	
1501	Float	W	Total DC Output Power	
6000	Float	W	Battery Power	
6001	Enum		Battery Charge/Discharge State	1000: Static 1001: Charging 1002: Discharging
6002	Float	%	Battery SOC	

6105	Float	%	Emergency power supply	
6004	Float	kWh	Battery Daily Charging Energy	
6005	Float	kWh	Battery Daily Discharging Energy	
6006	Float	kWh	Battery Total Charging Energy	
6007	Float	kWh	Battery Total Discharging Energy	
7120	Enum		Meter Connection Status	1000: Enable 1001: Disable
21028	Float	W	Meter Power	

- **INDEVOLT SolidFlex2000/PowerFlex2000**

cJSON Point	cJSON Value Type	Unit	Point Description	Enum Definition
SN				
0	String		Device SN	
Firmware Version Information				
1118	String		PG2000Series EMS	
1109	String		PG2000Series BMS-MB	
1119	String		PG2000Series PCS	
1120	String		PG2000Series DCDC	
1136	String		SFA/PFA DCDC1	
1137	String		SFA/PFA BMS1	

1138	String		SFA/PFA DCDC2	
1139	String		SFA/PFA BMS2	
1140	String		SFA/PFA DCDC3	
1141	String		SFA/PFA BMS3	
1142	String		SFA/PFA DCDC4	
1143	String		SFA/PFA BMS4	
1098	String		SFA/PFA DCDC5	
1099	String		SFA/PFA BMS5	
System Operating Information				
7101	Enum		Working Mode	1: Self-consumed Prioritized 4: Real-time Control 5: Charge/Discharge Schedule
142	Num	kWh	Rated Capacity	
6105	Num	%	Emergency Power Supply	
2618	Num		Grid Charging	1000: Disable 1001: Enable
11009	Num	W	Inverter Input Limit	
2101	Num	W	Total AC lutput Power	
2108	Num	W	Total AC Output Power	
11010	Num	W	Feed-in Power Limit	

2108	Num	W	Max AC Output Power	
Bypass Power				
667	Float	W	Bypass power	
Electrical Energy Information				
2107	Num	kWh	Total AC Output Energy	
2104	Num	Wh	Total AC Input Energy	
2105	Num	kWh	Off-grid Output Energy	
11034	Num	Wh	Bypass Input Energy	
1502	Num	kWh	Daily PV Generation	
6004	Num	kWh	Battery Daily Charging Energy	
6005	Num	kWh	Battery Daily Discharging Energy	
6006	Num	kWh	Battery Total Charging Energy	
6007	Num	kWh	Battery Total Discharging Energy	
Electricity Meter Status				
7120	Enum	W	Meter Connection Status	1000: Enable 1001: Disable
11016	Float	W	Meter Power	
Grid Information				

2600	Num	V	Grid Voltage	
2612	Num	Hz	Grid Frequency	
Battery Pack Operating Parameters				
6001	Enum		Battery Charge/Discharge State	1000: Static 1001: Charging 1002: Discharging
6000	Num	W	Battery Power	
6002	Num	%	Battery SOC Total	
9008	String		Batt SN-MB	
9000	Num	%	Batt SOC-MB	
9004	Num	V	Batt V-MB	
9013	Num	A	Batt I-MB	
9012	Num	°C	Batt Temp-MB	
9032	String		Batt SN-Slave1	
9016	Num	%	Batt SOC-Slave1	
9020	Num	V	Batt V-Slave1	
19173	Num	A	Batt I-Slave1	
9030	Num	°C	Batt Temp-Slave1	
9051	String		Batt SN-Slave2	
9035	Num	%	Batt SOC-Slave2	
9039	Num	V	Batt V-Slave2	
19174	Num	A	Batt I-Slave2	
9049	Num	°C	Batt Temp-Slave2	
9070	String		Batt SN-Slave3	

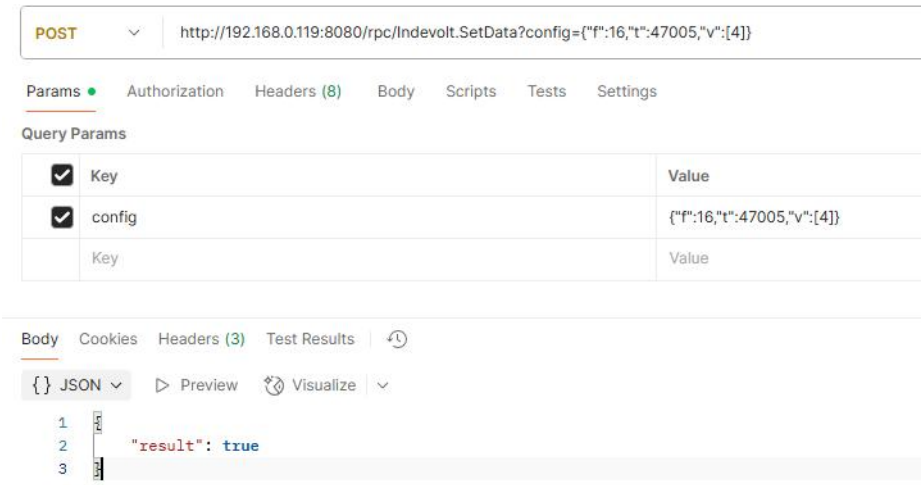
9054	Num	%	Batt SOC-Slave3	
9058	Num	V	Batt V-Slave3	
19175	Num	A	Batt I-Slave3	
9068	Num	°C	Batt Temp-Slave3	
9165	Num		Batt SN-Slave4	
9149	Num	%	Batt SOC-Slave4	
9153	Num	V	Batt V-Slave4	
19176	Num	A	Batt I-Slave4	
9163	Num	°C	Batt Temp-Slave4	
9218	String		Batt SN-Slave5	
9202	Num	%	Batt SOC-Slave5	
9206	Num	V	Batt V-Slave5	
19177	Num	A	Batt I-Slave5	
9216	Num	°C	Batt Temp-Slave5	
PV Operating Parameters				
1501	Num	w	Total DC Output Power	
1632	Num	A	DC Input Current 1	
1600	Num	V	DC Input Voltage 1	
1664	Num	W	DC Input Power 1	
1633	Num	A	DC Input Current 2	
1601	Num	V	DC Input Voltage 2	
1665	Num	W	DC Input Power 2	
1634	Num	A	DC Input Current 3	
1602	Num	V	DC Input Voltage 3	

1666	Num	W	DC Input Power 3	
1635	Num	A	DC Input Current 4	
1603	Num	V	DC Input Voltage 4	
1667	Num	W	DC Input Power 4	

2. Control device

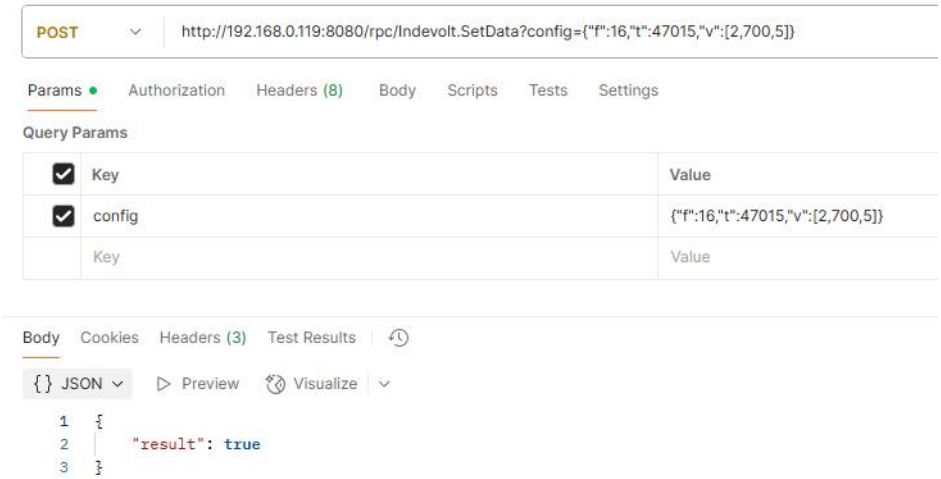
(1) Set working mode

Request example	**POST** http://192.168.31.213:8080/rpc/Indevolt.SetData?config={"f":16,"t":47005,"v":[4]}
Response example	{"result": true}



(2) Set discharging state, power and SOC.

Request example	POST http://192.168.31.213:8080/rpc/Indevolt.SetData?config={"f":16,"t":47015,"v":[2,700,5]}
Response example	{"result": true}



Function: Indevolt.SetData

Parameter

Parameter	Type	Description
config	Object	f: function code, 16 by default; t: register address, see the **cJSON Point** column in the table below; v: write value, see the **Value** column in the table below.

- **INDEVOLT BK1600/BK1600Ultra**

cJson Point	cJson Value Type	Unit	Point Description	Value
47005	Enum		Mode	1: Self-consumed Prioritized 2: Charge/discharge Schedule 4: Real-time Control
47015	UINT		State	0: Standby 1: Charging 2: Discharging

47016	INT	w	Power	Charging: 0-1200 Discharging: 0-800
47017	UINT	%	SOC	0-100

- **INDEVOLT SolidFlex2000/PowerFlex2000**

cJSON Point	cJSON Value Type	Unit	Point Description	Value
47005	Enum		Mode	1: Self-consumed Prioritized 2: Charge/Discharge Schedule 4: Real-time control
47015	UINT		State	0: Standby 1: Charging 2: Discharging
47016	INT	W	Power	MAX Charging: 50–2400 MAX Discharging: 50–2400
47017	UINT	%	SOC	5-100
1147	INT	W	Max AC Output Power	
1146	INT	W	Feed-in Power Limit	
1143	UINT		Grid Charging	0: Disable 1: Enable
1138	INT	W	Inverter Input Limit	
7266	Enum		Bypass	0: Disable 1: Enable

1142	INT	%	Backup SOC	
7265	Enum		Light	0: Disable 1: Enable

Note: In real-time control mode, device charging/discharging can be controlled by simultaneously writing state, power, and SOC values.

Return value

Parameter	Type	Description
result	Bool	true: success; false: failure.

Sys

1. Obtain CMS (Communication Management System) information

请求示例	**GET** http://192.168.31.213:8080/rpc/Sys.GetConfig

返回值示例	{
	"device": {
	"hostname": "",
	"timezone": 480,
	"type": "CMS-SF2000",
	"sn": "",
	"mac": "7C3E82EF997F",
	"fw": "T1.4.06_ROOD. 072_M4801_0000002C",
	"f_ver": "T1406.07.002C",
	"p_ver": "VOD.00.08",
	"time": "2025-12-18 09:44:57",
	"time_stamp": 1766051097,

	"run_time": 2244 } }

Function: Sys.GetConfig

Return value

Parameter	Type	Description
device	Object	See the table below for device configuration information.

device configuration

Parameter	Type	Description
hostname	String	Device name
timezone	Num	Timezone
type	String	Device model
sn	String	Device serial number
mac	String	Device MAC address
fw	String	Device firmware version
f_ver	String	PG2000Series CMS
p_ver	string	PG2000Series Pfile
time	String	Current time
time_stamp	Num	Current timestamp (in seconds)
run_time	Num	Device runtime (in seconds)

5. FAQ

Q: HTTP request returns 401 Unauthorized。

A:

- Verify the username and password for Digest Authentication.
- New/factory-reset devices only support access to the designated interface: User.SetConfig. Refer to Digest Authentication section - other APIs become accessible after successful password modification and authentication with the new password.

Q: Device fails to return IP address after broadcast command.

A: The OpenData API is not enabled, making this function inactive. Refer to Enable API section.